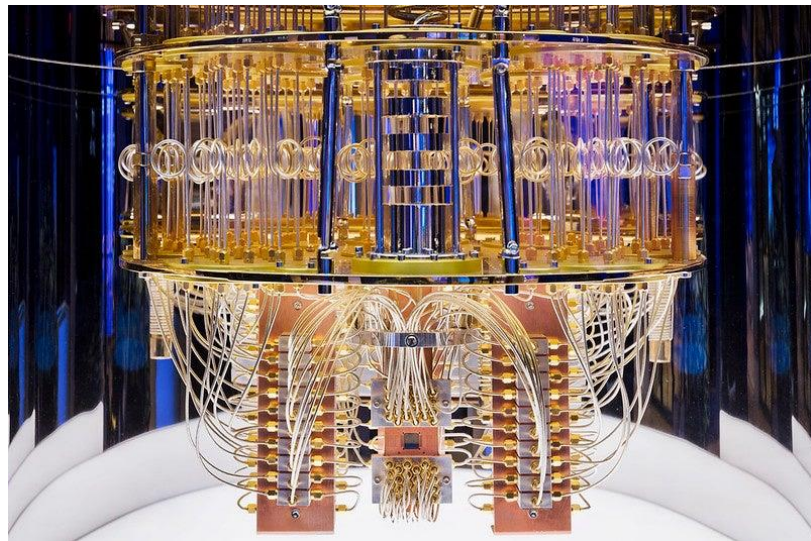


Kawasaki Quantum Summer Camp 2025

IBM Quantumの アカウントの設定

Jul 30, 2025

沼田祈史
Kifumi Numata
IBM Quantum



IBM Quantum

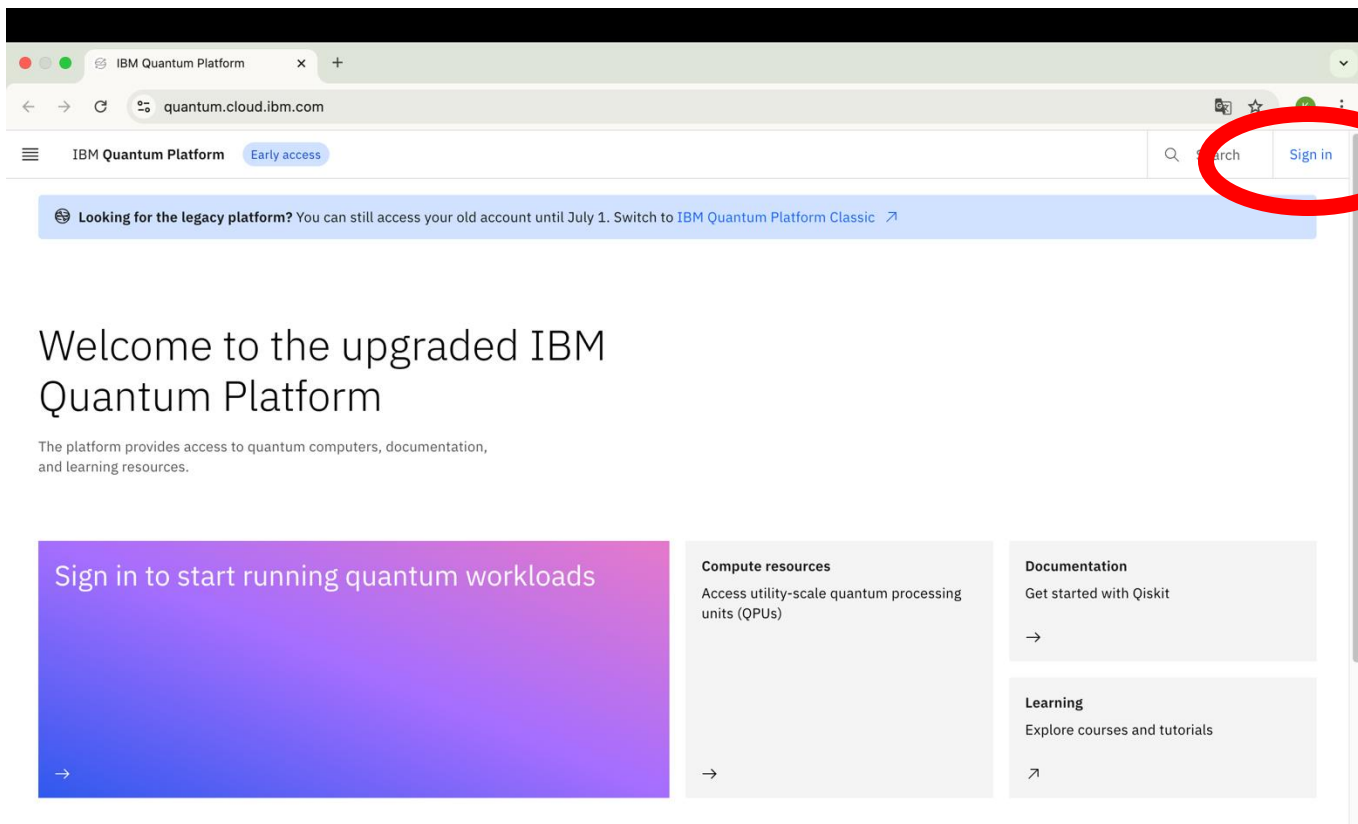
**本物の量子コンピューターにインターネット経由で計算
をさせる時に必要になるアカウントを作成します！**

もくじ

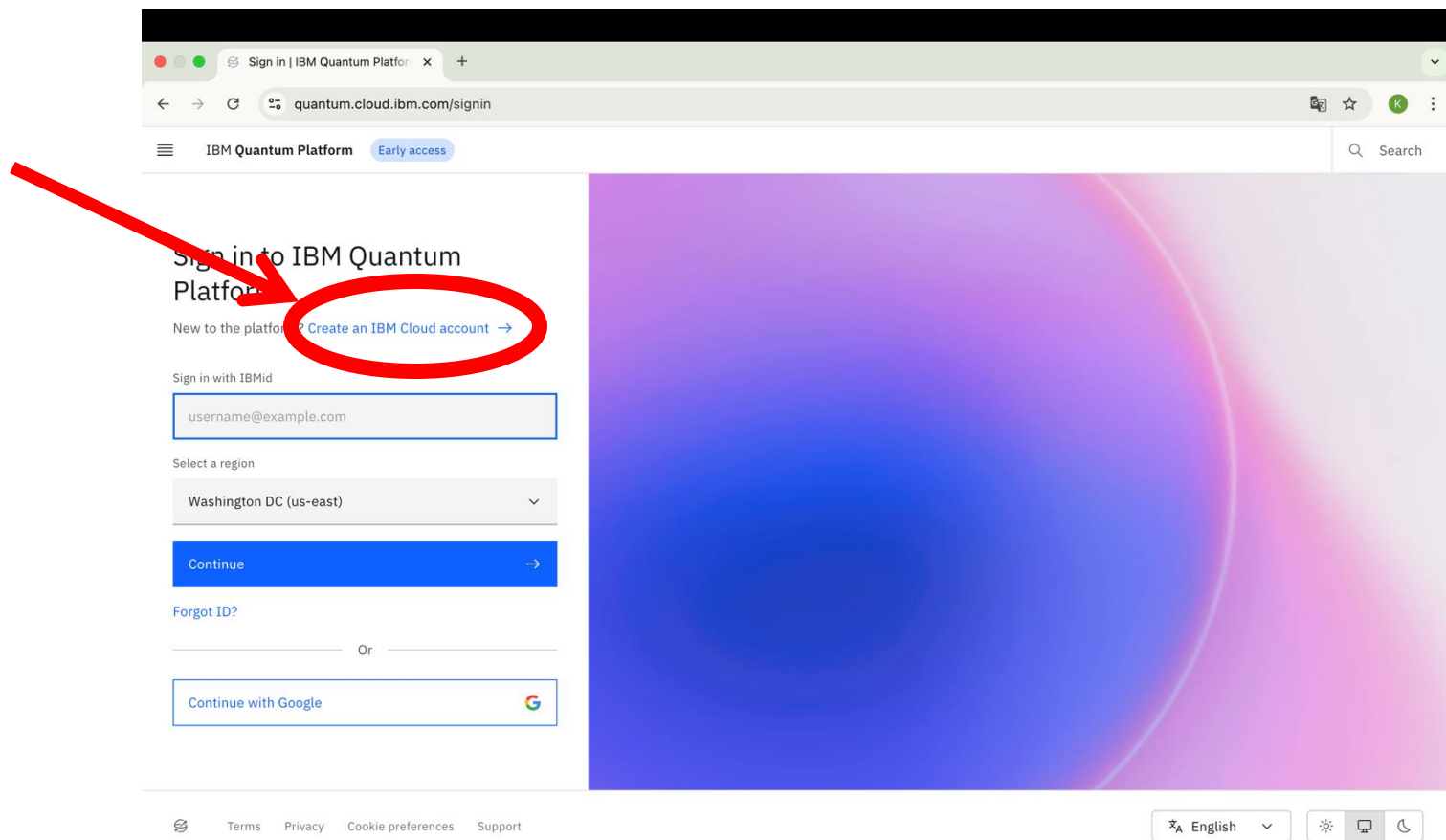
- 1. Feature code で IBM Cloud account を作る**
- 2. Instance を作り、CRNを得る**
- 3. API key をダウンロードする**

1. Feature codeでIBM Cloud accountを作る

ウェブブラウザで <https://quantum.cloud.ibm.com/> にいき、「Sign in」をクリックします。



「Create an IBM Cloud account」をクリックします。



「Eメール」と「パスワード」を入力し、「Eメールの検証」をクリックします。

The screenshot shows a web browser window with the URL `cloud.ibm.com/registration?cm_sp=quantum&utm_content=quantum-next&target=https%3A%2F%2Fquantum.cloud.ibm.com&error_uri=`. The page title is "Create an IBM Cloud account". Below the title, there is a link: "既に IBM Cloud アカウントをお持ちの場合 ログイン". The main form is titled "アカウント情報" (Account Information) and contains the following fields and buttons:

- Eメール** (Email): A text input field with a blue arrow icon on the right.
- パスワード** (Password): A password input field with a toggle icon on the right.
- OR**: A separator between the password field and the Google sign-up button.
- Sign up with Google**: A button with the Google logo.
- 次のステップ ↓** (Next Step): A button with a downward arrow.

Below the main form, there are two sections with radio buttons:

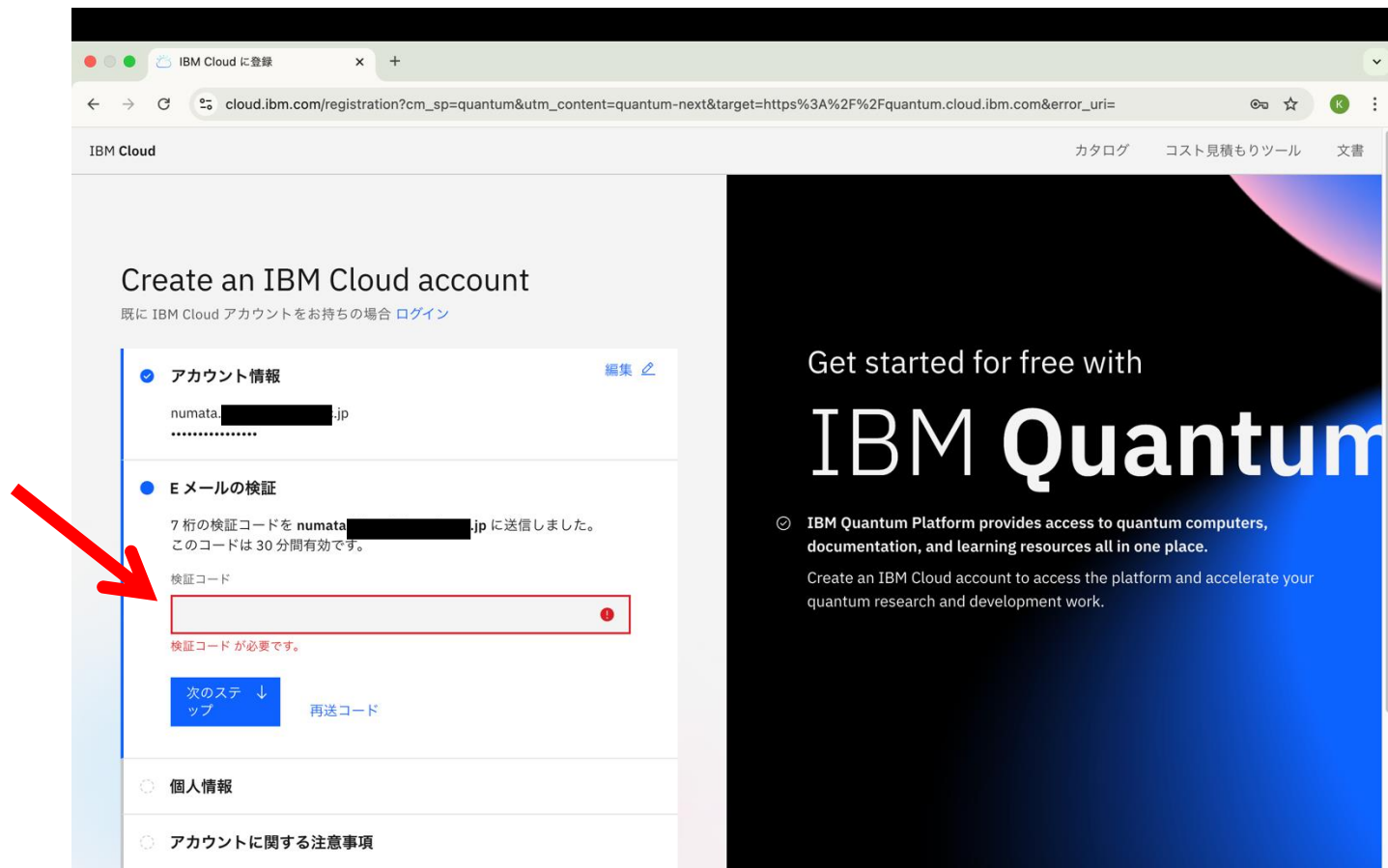
- Eメールの検証** (Email Verification): This option is selected.
- 個人情報** (Personal Information): An unselected option.

On the right side of the page, there is a large banner for "Get started for free with IBM Quantum". Below the banner, there is a list of features:

- IBM Quantum Platform provides access to quantum computers, documentation, and learning resources all in one place.

Below the list, there is a text: "Create an IBM Cloud account to access the platform and accelerate your quantum research and development work." On the far right, there is a vertical button labeled "Cookieの設定" (Cookie Settings).

メールを確認し、「検証コード」を入力します。



The screenshot shows a web browser window with the URL `cloud.ibm.com/registration?cm_sp=quantum&utm_content=quantum-next&target=https%3A%2F%2Fquantum.cloud.ibm.com&error_uri=`. The page title is "Create an IBM Cloud account". Below the title, there is a link for users who already have an account: "既に IBM Cloud アカウントをお持ちの場合 ログイン". The registration form is divided into sections: "アカウント情報" (Account Information), "Eメールの検証" (Email Verification), "個人情報" (Personal Information), and "アカウントに関する注意事項" (Important information about the account). The "Eメールの検証" section is currently active, showing that a 7-digit verification code has been sent to the email address `numata. [redacted] .jp` and is valid for 30 minutes. A red arrow points to the "検証コード" (Verification Code) input field, which is currently empty. Below the input field, a red error message states "検証コード が必要です。" (Verification code is required.). To the right of the input field are two buttons: "次のステップ ↓" (Next Step) and "再送コード" (Resend Code). The right side of the page features a large banner for "IBM Quantum" with the text "Get started for free with IBM Quantum" and a list of benefits: "IBM Quantum Platform provides access to quantum computers, documentation, and learning resources all in one place." and "Create an IBM Cloud account to access the platform and accelerate your quantum research and development work."

Create an IBM Cloud account

既に IBM Cloud アカウントをお持ちの場合 [ログイン](#)

アカウント情報 [編集](#)

numata. [redacted] .jp

Eメールの検証

7桁の検証コードを numata. [redacted] .jp に送信しました。
このコードは 30 分間有効です。

検証コード

検証コード が必要です。

次のステップ ↓

再送コード

個人情報

アカウントに関する注意事項

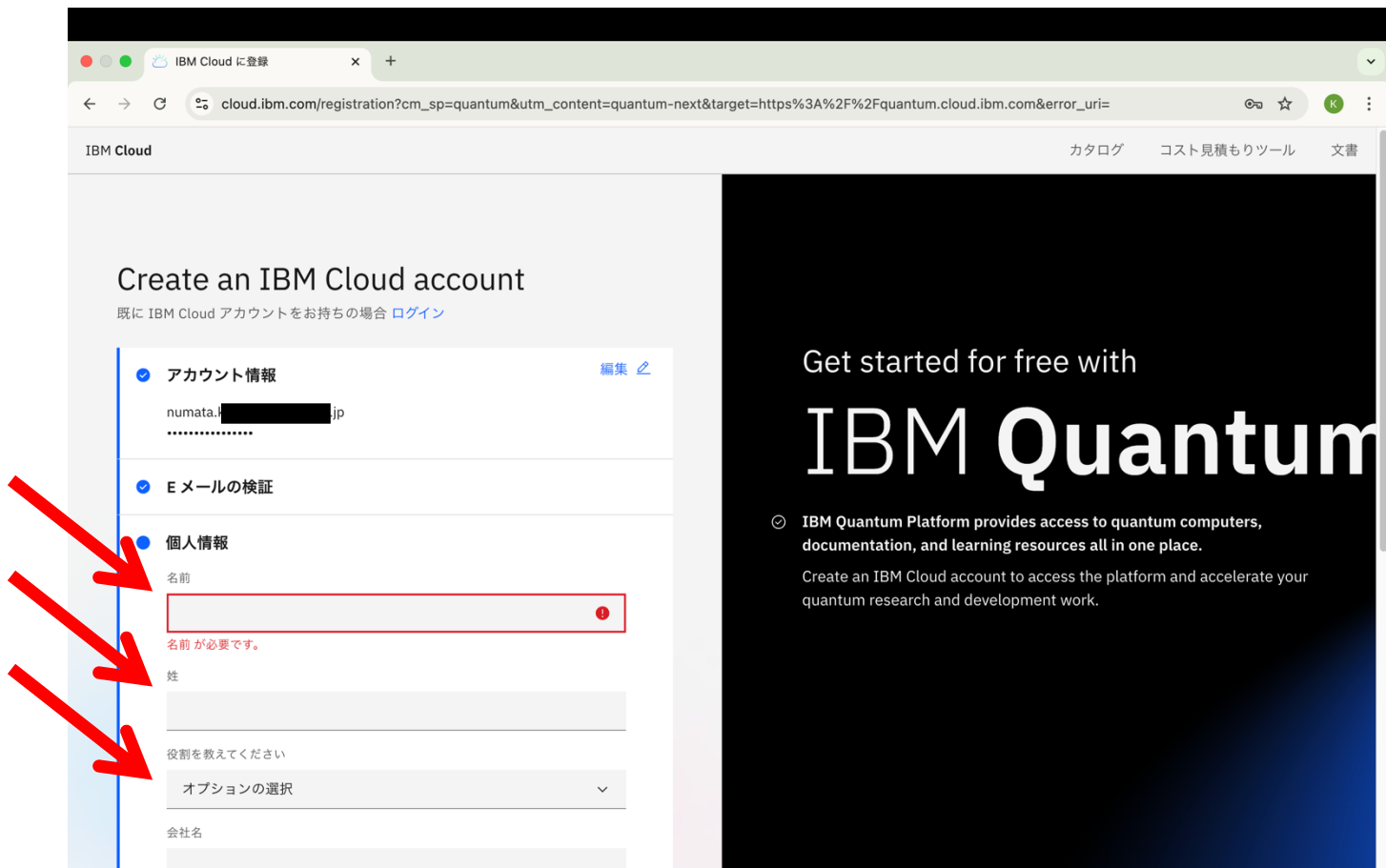
Get started for free with

IBM Quantum

- IBM Quantum Platform provides access to quantum computers, documentation, and learning resources all in one place.

Create an IBM Cloud account to access the platform and accelerate your quantum research and development work.

「名前」、「姓」などを入力します。



The screenshot shows the 'Create an IBM Cloud account' page. The 'Personal Information' section is highlighted with a blue dot and a red vertical line. Three red arrows point to the '名前' (Name), '姓' (Surname), and '役割を教えてください' (Tell us your role) fields. The '名前' field has a red border and an error message '名前が必要です。' (Name is required.).

IBM Cloud

カタログ コスト見積もりツール 文書

Create an IBM Cloud account

既に IBM Cloud アカウントをお持ちの場合 [ログイン](#)

- ✓ アカウント情報 [編集](#)
numata.1[REDACTED].jp
- ✓ Eメールの検証
- 個人情報

名前

姓

役割を教えてください

オプションの選択

会社名

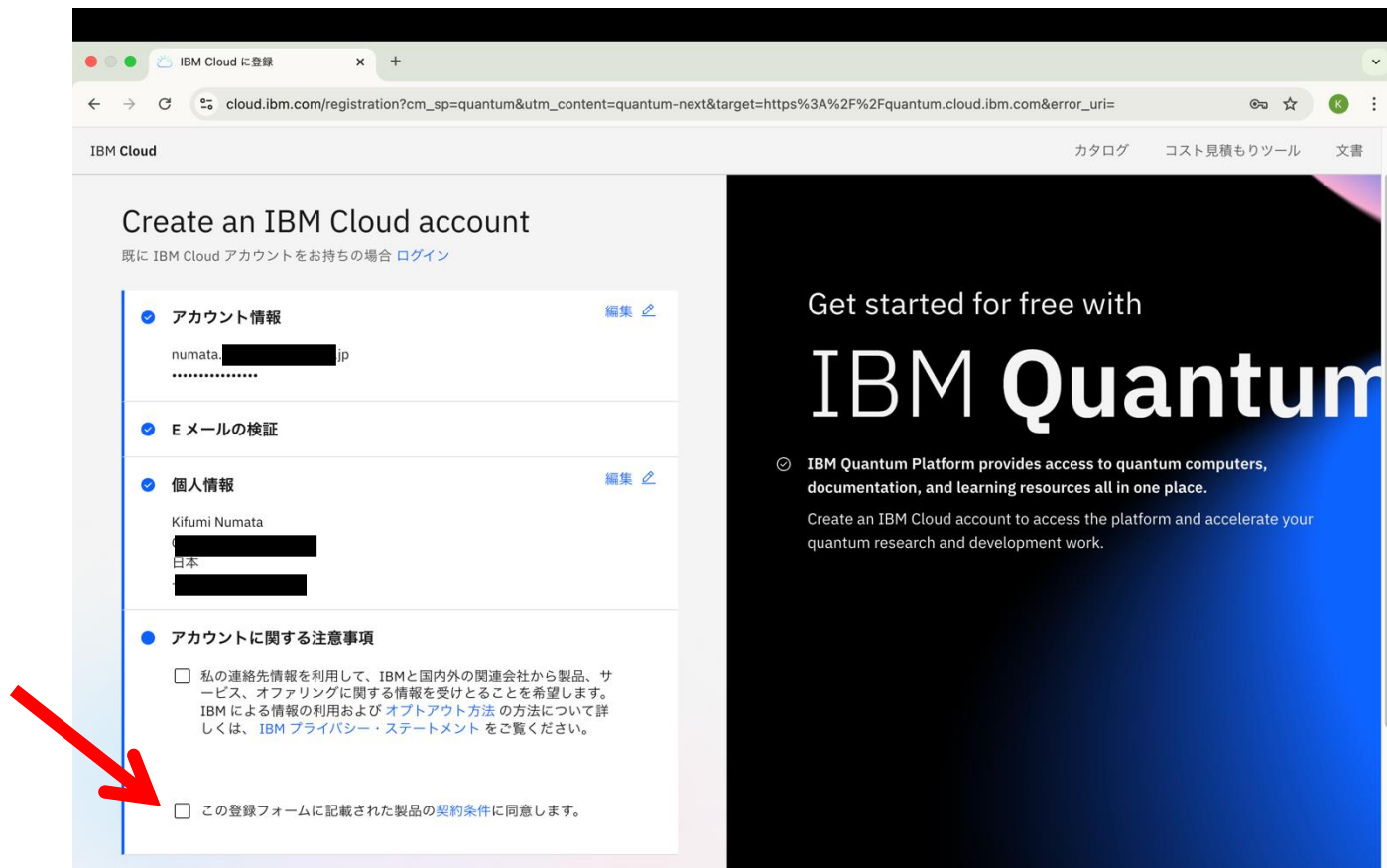
Get started for free with

IBM Quantum

- ◎ IBM Quantum Platform provides access to quantum computers, documentation, and learning resources all in one place.

Create an IBM Cloud account to access the platform and accelerate your quantum research and development work.

「アカウントに関する注意事項」を確認してチェックします。

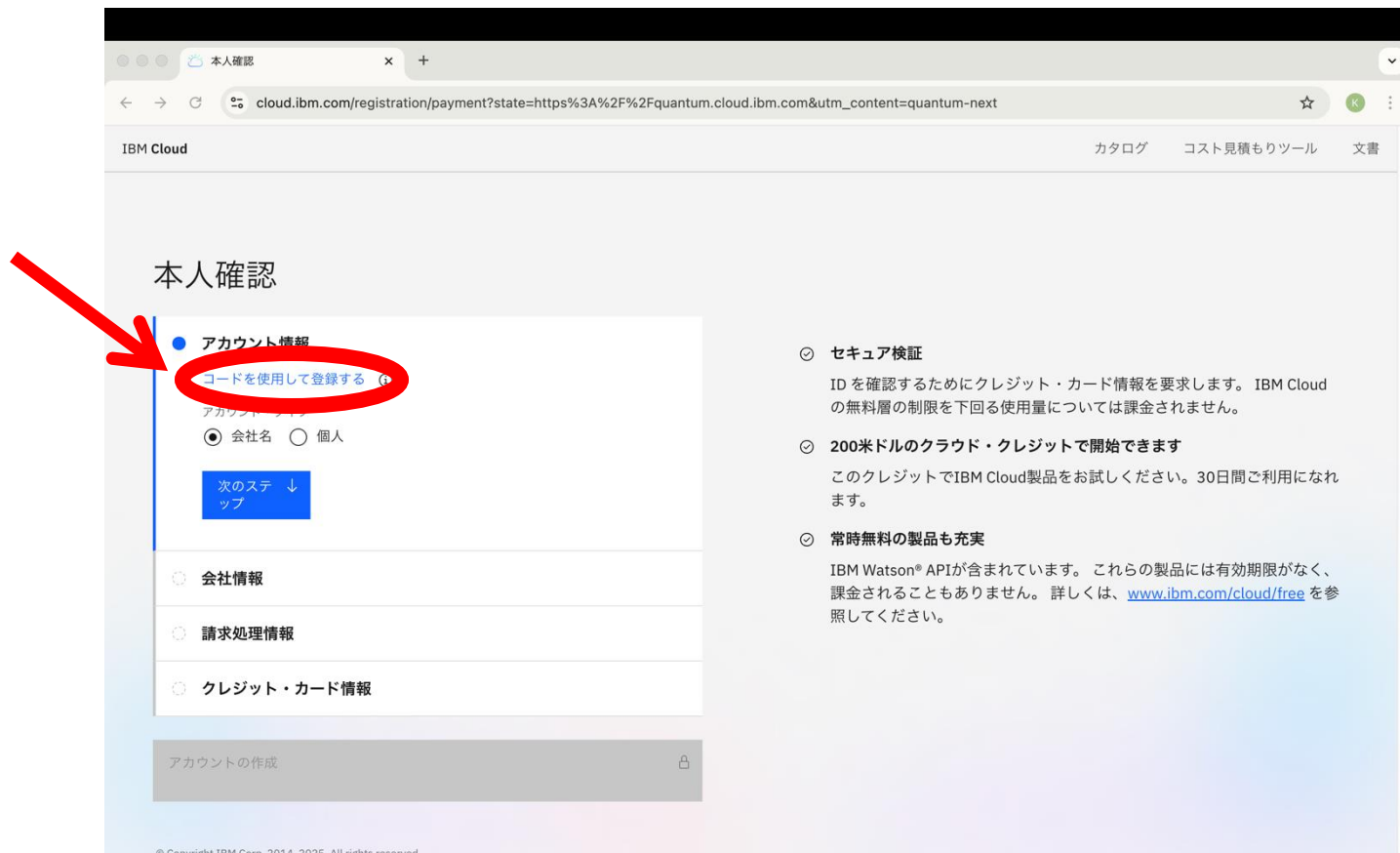


The screenshot shows the IBM Cloud registration page. The browser address bar displays the URL: `cloud.ibm.com/registration?cm_sp=quantum&utm_content=quantum-next&target=https%3A%2F%2Fquantum.cloud.ibm.com&error_uri=`. The page title is "Create an IBM Cloud account". Below the title, there is a link for existing users: "既に IBM Cloud アカウントをお持ちの場合 ログイン". The registration form is divided into four sections, each with a blue checkmark icon and a "編集" (Edit) link:

- アカウント情報** (Account Information): Includes a field for "numata. [redacted] jp".
- Eメールの検証** (Email Verification): No visible input fields.
- 個人情報** (Personal Information): Includes fields for "Kifumi Numata", a phone number "[redacted]", and "日本" (Japan).
- アカウントに関する注意事項** (Terms and Conditions): This section contains two checkboxes. The first checkbox is unchecked and is highlighted by a red arrow. The text next to it reads: "私の連絡先情報を利用して、IBMと国内外の関連会社から製品、サービス、オファリングに関する情報を受けとることを希望します。IBM による情報の利用および [オプトアウト方法](#) の方法について詳しくは、[IBM プライバシー・ステートメント](#) をご覧ください。"

On the right side of the page, there is a large banner with the text "Get started for free with IBM Quantum". Below this, it states: "IBM Quantum Platform provides access to quantum computers, documentation, and learning resources all in one place. Create an IBM Cloud account to access the platform and accelerate your quantum research and development work."

「コードを使用して登録する」をチェックします。



The screenshot shows the IBM Cloud registration page. A red arrow points to the '登録する' button in the 'アカウント情報' section. The page is titled '本人確認' (Self-Verification). The 'アカウント情報' section includes a dropdown menu with 'コードを使用して登録する' selected, radio buttons for '会社名' (selected) and '個人' (unselected), and a '次のステップ' button. Below this are sections for '会社情報', '請求処理情報', and 'クレジット・カード情報'. To the right, there are three informational points: 'セキュア検証', '200米ドルのクラウド・クレジットで開始できます', and '常時無料の製品も充実'.

IBM Cloud

カタログ コスト見積もりツール 文書

本人確認

● アカウント情報

コードを使用して登録する (i)

会社名 個人

次のステップ ↓

○ 会社情報

○ 請求処理情報

○ クレジット・カード情報

アカウントの作成

○ セキュア検証

IDを確認するためにクレジット・カード情報を要求します。IBM Cloudの無料層の制限を下回る使用量については課金されません。

○ 200米ドルのクラウド・クレジットで開始できます

このクレジットでIBM Cloud製品をお試しください。30日間ご利用になれます。

○ 常時無料の製品も充実

IBM Watson® APIが含まれています。これらの製品には有効期限がなく、課金されることもありません。詳しくは、www.ibm.com/cloud/freeを参照してください。

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メールを確認し、「Kayla Lee」からの「Start your journey on upgraded IQP」というのメールの中からfeature codeをコピーします。

Start your journey on upgraded IQP

外部

受信トレイ x



Kayla Lee <Kayla.Lee@ibm.com>

2025/05/15 1:15



To 自分 ▼



日本語に翻訳



Hi Kifumi Numata,

Thanks for being a Qiskit Advocate! As a special member of the community, we're providing you with a feature code to extend the length of your trial account on IBM Cloud.

Your assigned feature code is 36b5[REDACTED], which should be applied at time of registration using the instructions below. Feature codes are unique alphanumeric codes that can only be applied once and only applied to new accounts. If you already have an account on IBM Cloud, you cannot use this feature code.

To get started with your code, create an account at <https://quantum.cloud.ibm.com/> and select "Register with a code." You can then enter in your feature code and follow the rest of the instructions. You can see how many days are left in the trial by going to Manage > Account > Account Settings at the top of the screen.

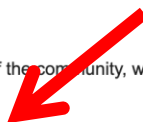
If you are having any problems applying the feature code, please respond to this message and we will try to help.

Enjoy your quantum access!

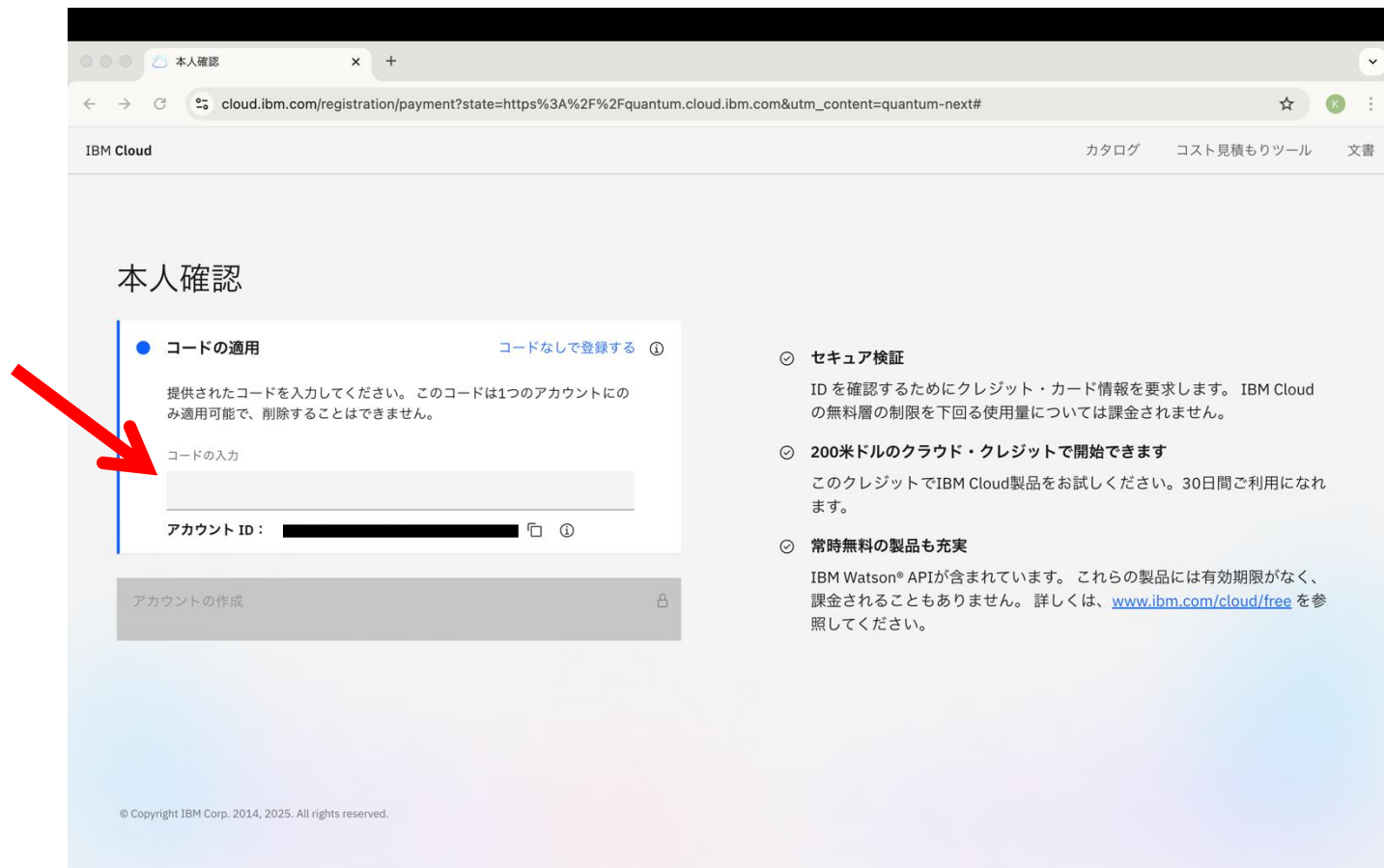
Best,

The team at IBM Quantum

この英数字をコピーします。



コピーしたfeature codeを「コードの入力」に貼り付けします。



cloud.ibm.com/registration/payment?state=https%3A%2F%2Fquantum.cloud.ibm.com&utm_content=quantum-next#

IBM Cloud

カタログ コスト見積もりツール 文書

本人確認

コードの適用 [コードなしで登録する](#) ⓘ

提供されたコードを入力してください。このコードは1つのアカウントにのみ適用可能で、削除することはできません。

コードの入力

アカウント ID: ⓘ

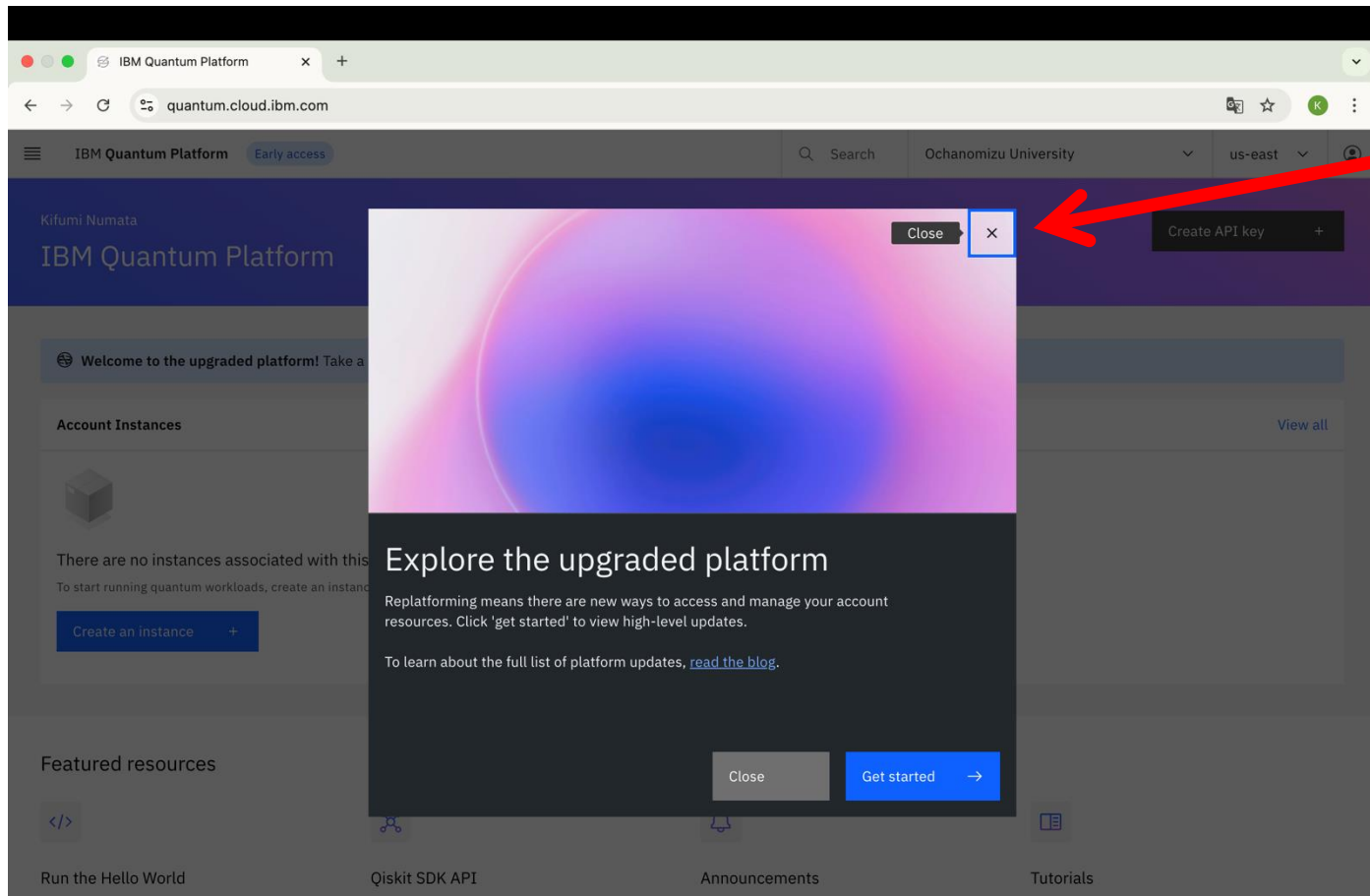
アカウントの作成

🔒 ⓘ

- 🔒 **セキュア検証**
IDを確認するためにクレジット・カード情報を要求します。IBM Cloudの無料層の制限を下回る使用量については課金されません。
- 🔒 **200米ドルのクラウド・クレジットで開始できます**
このクレジットでIBM Cloud製品をお試しください。30日間ご利用になれます。
- 🔒 **常時無料の製品も充実**
IBM Watson® APIが含まれています。これらの製品には有効期限がなく、課金されることもありません。詳しくは、www.ibm.com/cloud/free を参照してください。

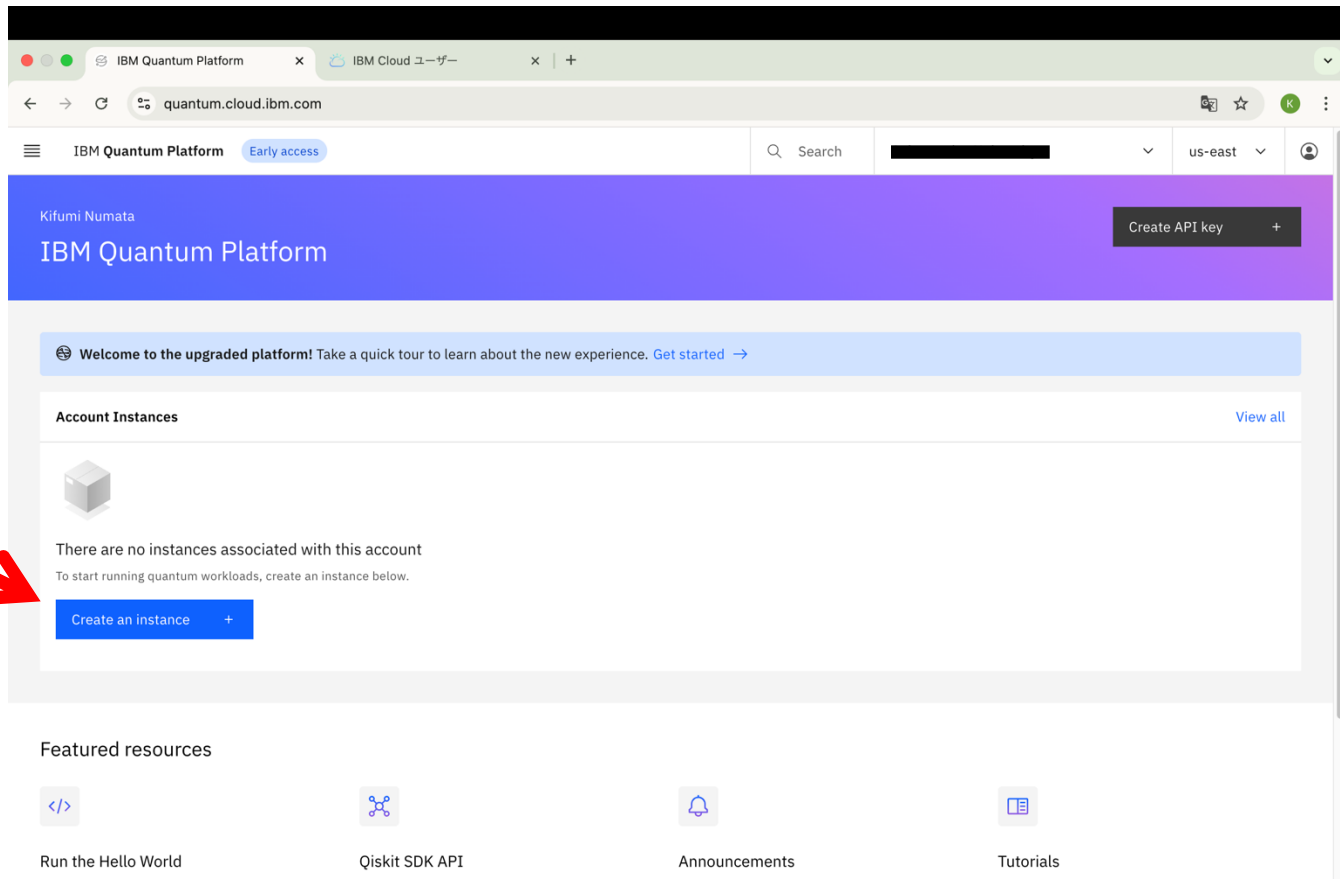
© Copyright IBM Corp. 2014, 2025. All rights reserved.

IBM Quantum Platformにログインできます。
ポップアップウィンドウを閉じるには右上「X」をクリックします。



2. Instance を作り、CRNを得る

「Create an Instance」をクリックします。



「Instance name」に好きな名前を入れます：「Open instance」など。

The screenshot shows the IBM Quantum Platform interface with a modal dialog titled "Create a new instance" for Ochanomizu University. The dialog has three tabs: "1. Details", "2. Compute resources", and "3. Access group". The "1. Details" tab is active. A red arrow points to the "Instance name" input field, which contains the placeholder text "Instance name". Below the input field, it states: "Allowed characters: letters, numbers, spaces, dots (.), underscores (_), colons (:), and hyphens (-). Max 256 characters." The "Resource group" is set to "Default". The "Region" is set to "Washington DC (us-east)". The "Tags (Optional)" field contains the example "env:dev, version-1". There are two pricing plans: "Open" (Free) and "Standard" (\$1.60 / second). The "Open" plan description says: "10 minutes free time per month on select IBM QPUs. After that limit is met, workloads will be blocked and no additional charges will occur." The "Standard" plan description says: "Pay-as-you-go for time used on select IBM QPUs. Maximum cost limits can be set per pay period." At the bottom of the dialog are "Cancel" and "Next" buttons.

IBM Quantum Platform

Early access

Search

us-east

Create API key

Kifumi Numata

IBM Quantum Platform

Welcome to the upgraded platform. To start running quantum workloads, create an instance.

Account Instances

Create an instance

There are no instances associated with this account.

Featured resources

Run the Hello World

Qiskit SDK API

Announcements

Tutorials

Ochanomizu University

Create a new instance

1. Details 2. Compute resources 3. Access group

Instance name

Resource group

Default

Allowed characters: letters, numbers, spaces, dots (.), underscores (_), colons (:), and hyphens (-). Max 256 characters.

Region

Washington DC (us-east)

Tags (Optional)

Example: env:dev, version-1

Select pricing plan

Open

Free

10 minutes free time per month on select IBM QPUs. After that limit is met, workloads will be blocked and no additional charges will occur.

Standard

\$1.60 / second

Pay-as-you-go for time used on select IBM QPUs. Maximum cost limits can be set per pay period.

View full details

Cancel

Next

「Open」を選択し、Termsを確認してチェックボックスをクリックします。

The screenshot shows the 'Create a new instance' dialog on the IBM Quantum Platform. The dialog has three tabs: '1. Details' (selected), '2. Compute resources', and '3. Access group'. Under '1. Details', there is a text input field for a name (placeholder: 'Max 256 characters.'), a 'Region' dropdown set to 'Washington DC (us-east)', and a 'Tags (Optional)' field with the example 'env:dev, version-1'. Below these is the 'Select pricing plan' section, which contains two options: 'Open' (Free, 10 minutes free time per month on select IBM QPUs. After that limit is met, workloads will be blocked and no additional charges will occur.) and 'Standard' (\$1.60 / second, Pay-as-you-go for time used on select IBM QPUs. Maximum cost limits can be set per pay period.). The 'Open' option is selected with a radio button. At the bottom of the dialog, there is a checkbox labeled 'I have read and agree to the following license agreements: Terms' which is also checked. The dialog has 'Cancel' and 'Next' buttons at the bottom.

IBM Quantum Platform

Create a new instance

1. Details 2. Compute resources 3. Access group

Max 256 characters.

Region Washington DC (us-east) Tags (Optional) Example: env:dev, version-1

Select pricing plan View full details

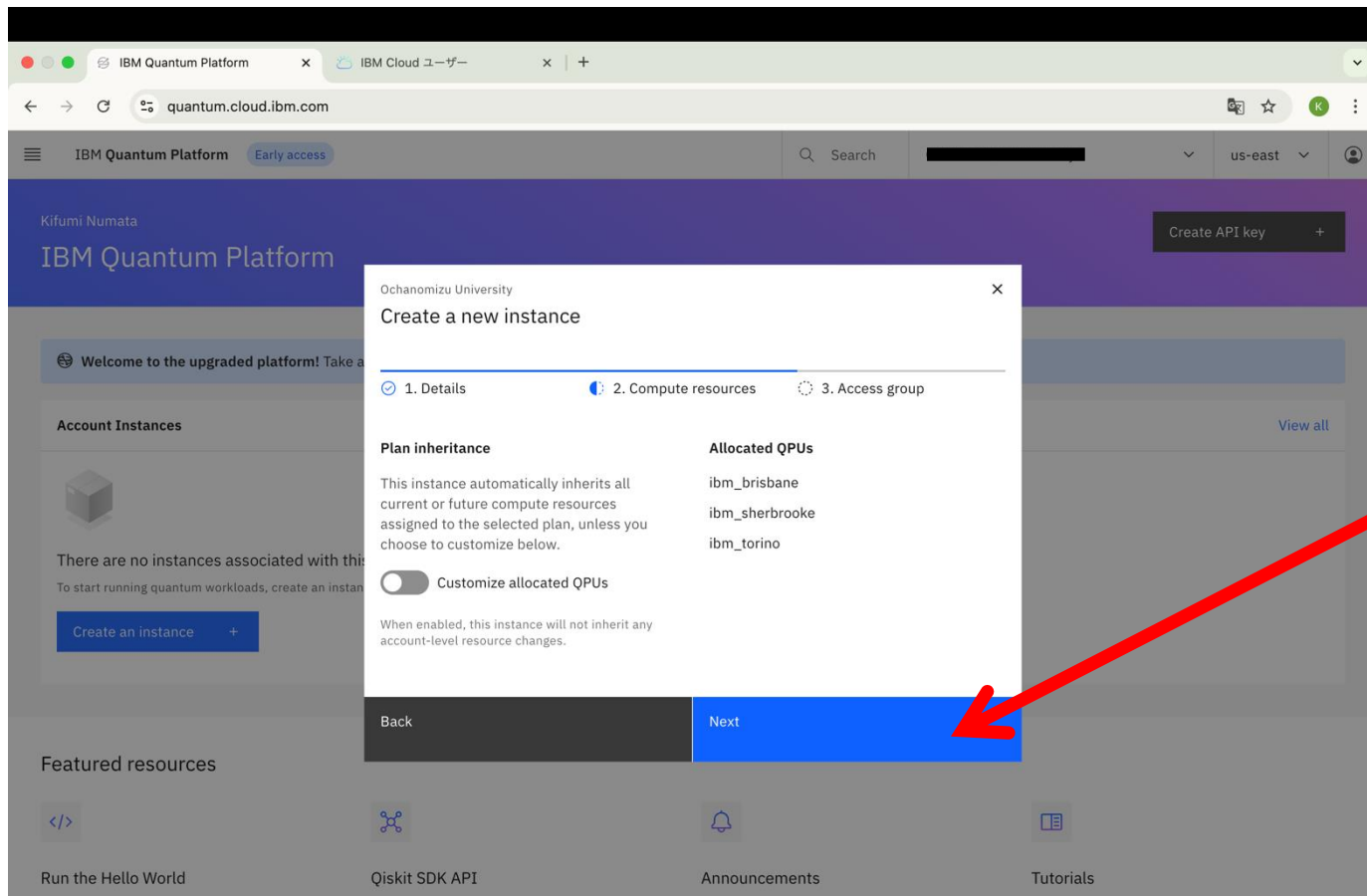
☒ Open Free 10 minutes free time per month on select IBM QPUs. After that limit is met, workloads will be blocked and no additional charges will occur.

☐ Standard \$1.60 / second Pay-as-you-go for time used on select IBM QPUs. Maximum cost limits can be set per pay period.

☒ I have read and agree to the following license agreements: Terms

Cancel Next

「Next」をクリックします。



「Create Instance」をクリックします。

The screenshot shows the IBM Quantum Platform interface with a modal dialog box titled "Create a new instance" for Ochanomizu University. The dialog has three steps: 1. Details, 2. Compute resources, and 3. Access group. It informs the user that an access group named "Open-kifumi" will be generated and lists its collaborators. A red arrow points to the "Create instance" button at the bottom right of the dialog.

IBM Quantum Platform

Create API key +

Account Instances

Create an instance +

Featured resources

Run the Hello World

Qiskit SDK API

Announcements

Tutorials

Create a new instance

1. Details 2. Compute resources 3. Access group

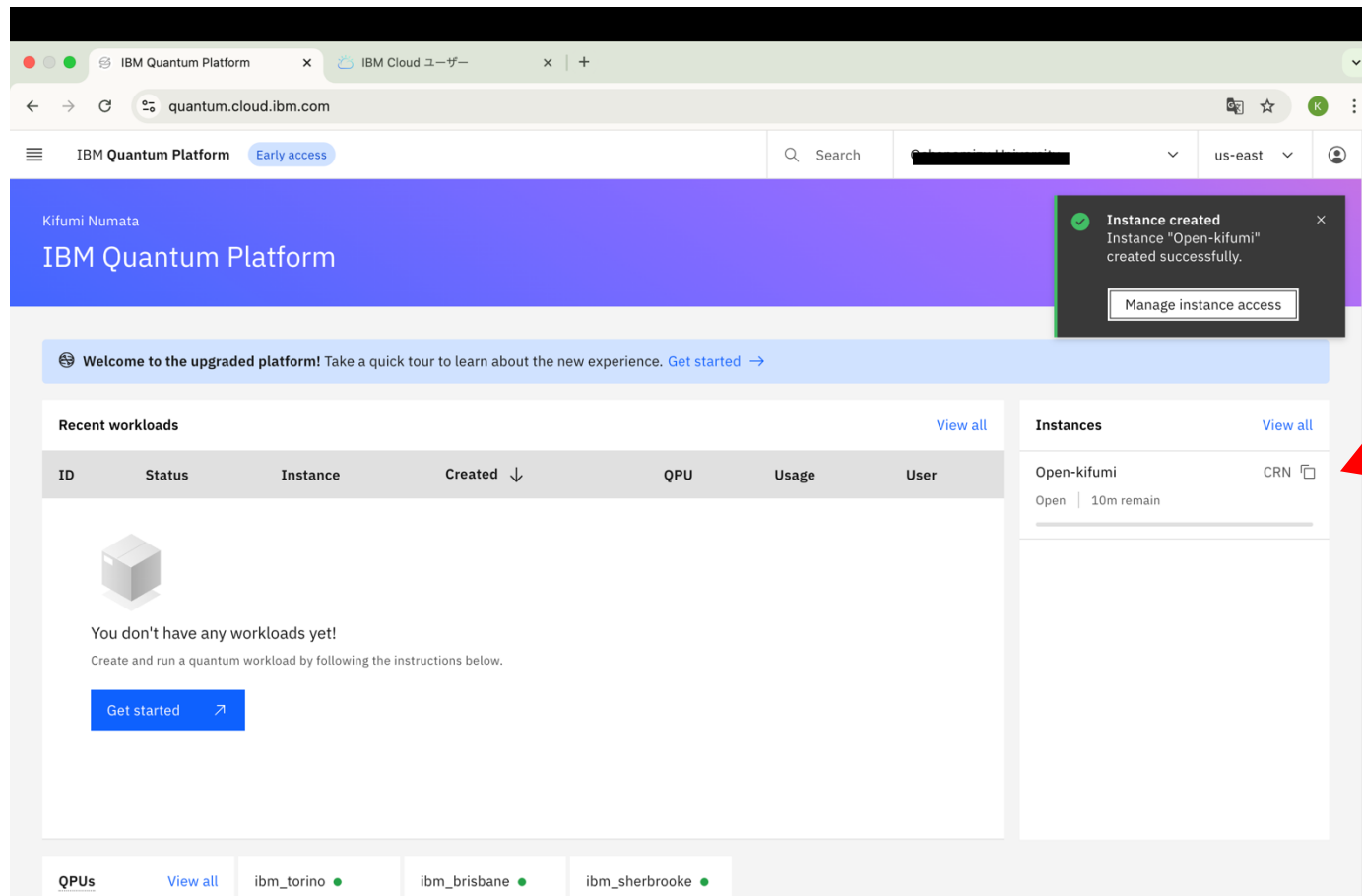
The following access group will be generated for "Open-kifumi":
Users must be added to this access group to run workloads on the instance. You can manage access group members from the users page.

Open-kifumi Collaborators
Collaborators can send workloads to QPUs using the time allocated to this instance.

Please note: You can manage access groups for this account in [IBM Cloud](#).

Back Create instance

Instanceが作れました。このトップページからCRNをコピーできます。



The screenshot shows the IBM Quantum Platform dashboard. A notification in the top right corner states: "Instance created Instance 'Open-kifumi' created successfully. Manage instance access". Below the notification, a red arrow points to the "CRN" column in the "Instances" table. The table lists one instance, "Open-kifumi", with a status of "Open" and "10m remain". The dashboard also features a "Recent workloads" section with a "Get started" button and a "QPU" section at the bottom.

IBM Quantum Platform

Welcome to the upgraded platform! Take a quick tour to learn about the new experience. [Get started](#)

Recent workloads [View all](#)

ID	Status	Instance	Created ↓	QPU	Usage	User
You don't have any workloads yet! Create and run a quantum workload by following the instructions below. Get started						

Instances [View all](#)

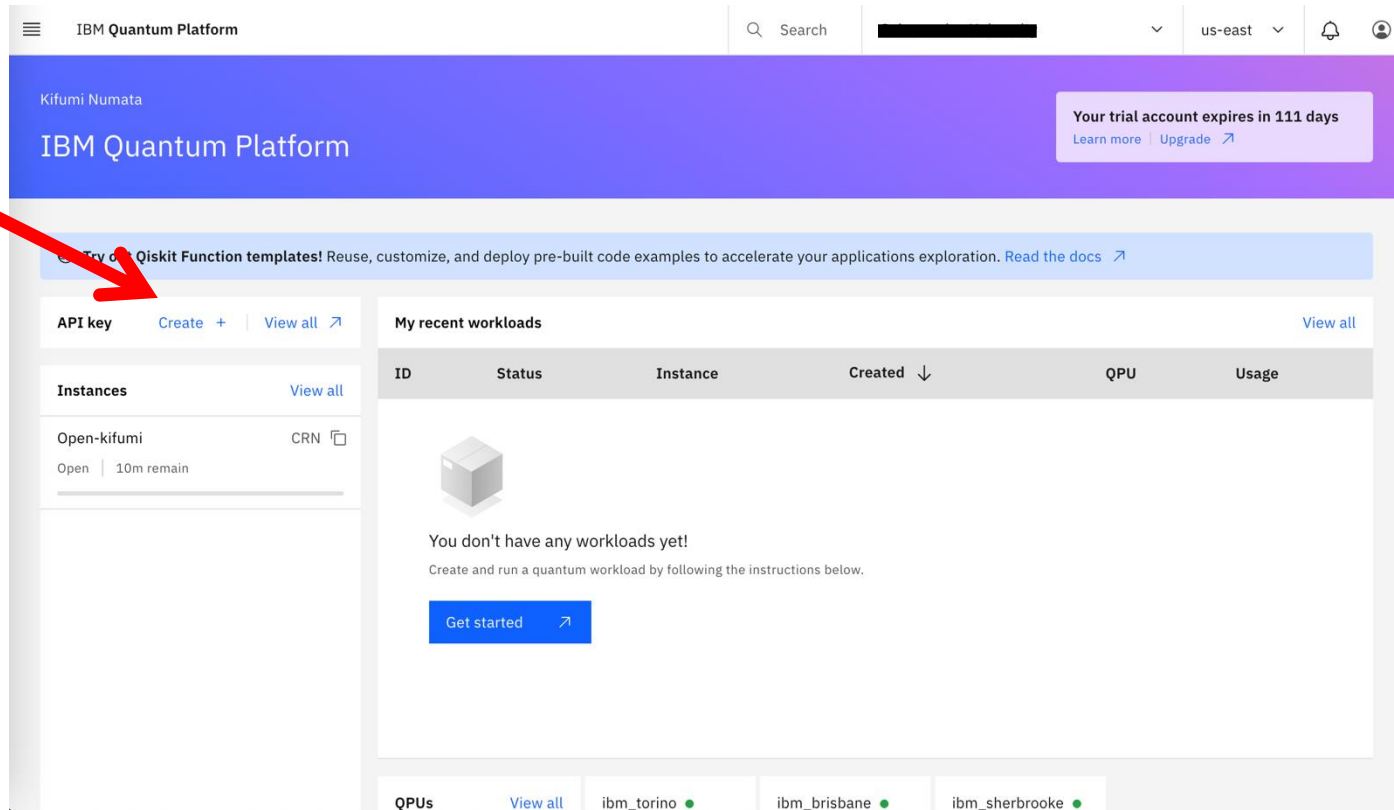
Instance	Status	CRN
Open-kifumi	Open 10m remain	CRN

QPU [View all](#)

- ibm_torino
- ibm_brisbane
- ibm_sherbrooke

3. API key をダウンロードする

API keyのとなりの「Create +」をクリックします。



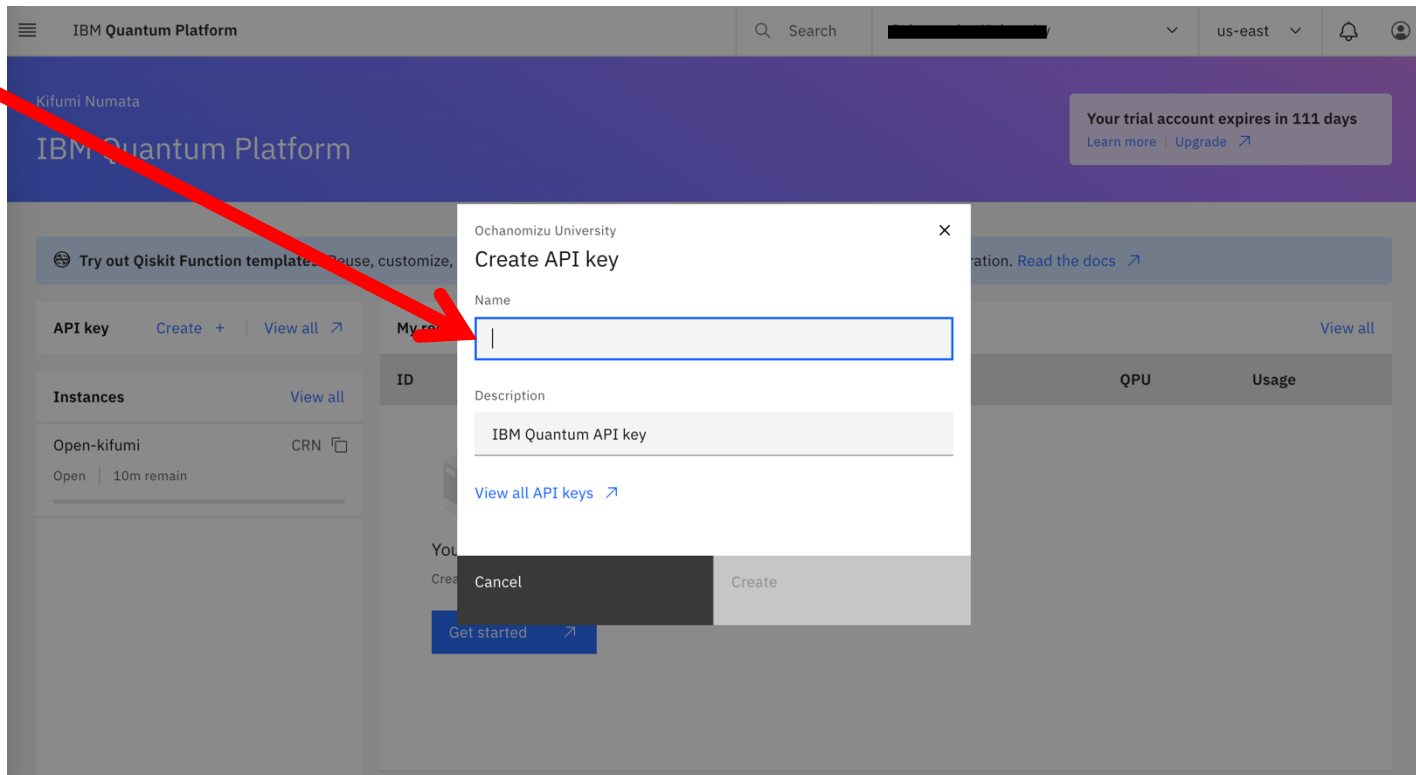
The screenshot shows the IBM Quantum Platform interface. At the top, there's a navigation bar with the IBM Quantum Platform logo, a search bar, and user information (Kifumi Numata). A blue banner below the navigation bar displays the user's name and a trial account expiration notice: "Your trial account expires in 111 days" with links for "Learn more" and "Upgrade".

Below the banner, there's a section for "API key" with a "Create +" button and a "View all" link. A red arrow points to the "Create +" button. Below this, there's a section for "Instances" with a "View all" link. The "Instances" section shows a table with one instance: "Open-kifumi" with a status of "Open" and "10m remain".

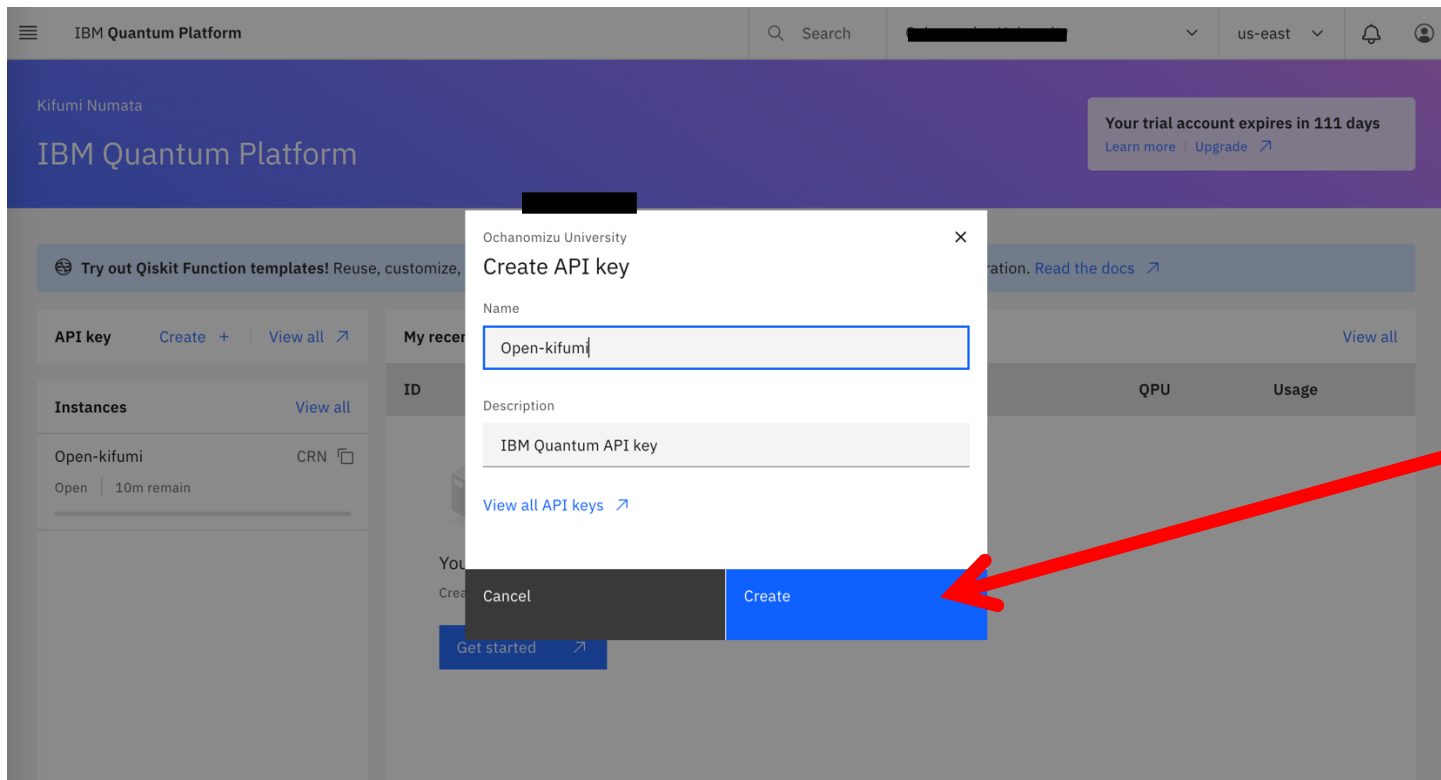
To the right of the "API key" and "Instances" sections, there's a section for "My recent workloads" with a "View all" link. This section contains a table with columns: "ID", "Status", "Instance", "Created", "QPU", and "Usage". Below the table, there's a message: "You don't have any workloads yet! Create and run a quantum workload by following the instructions below." with a "Get started" button.

At the bottom, there's a section for "QPUs" with a "View all" link. This section shows a table with columns: "QPU", "Status", and "Usage". The table lists three QPUs: "ibm_torino", "ibm_brisbane", and "ibm_sherbrooke", all with a status of "Online".

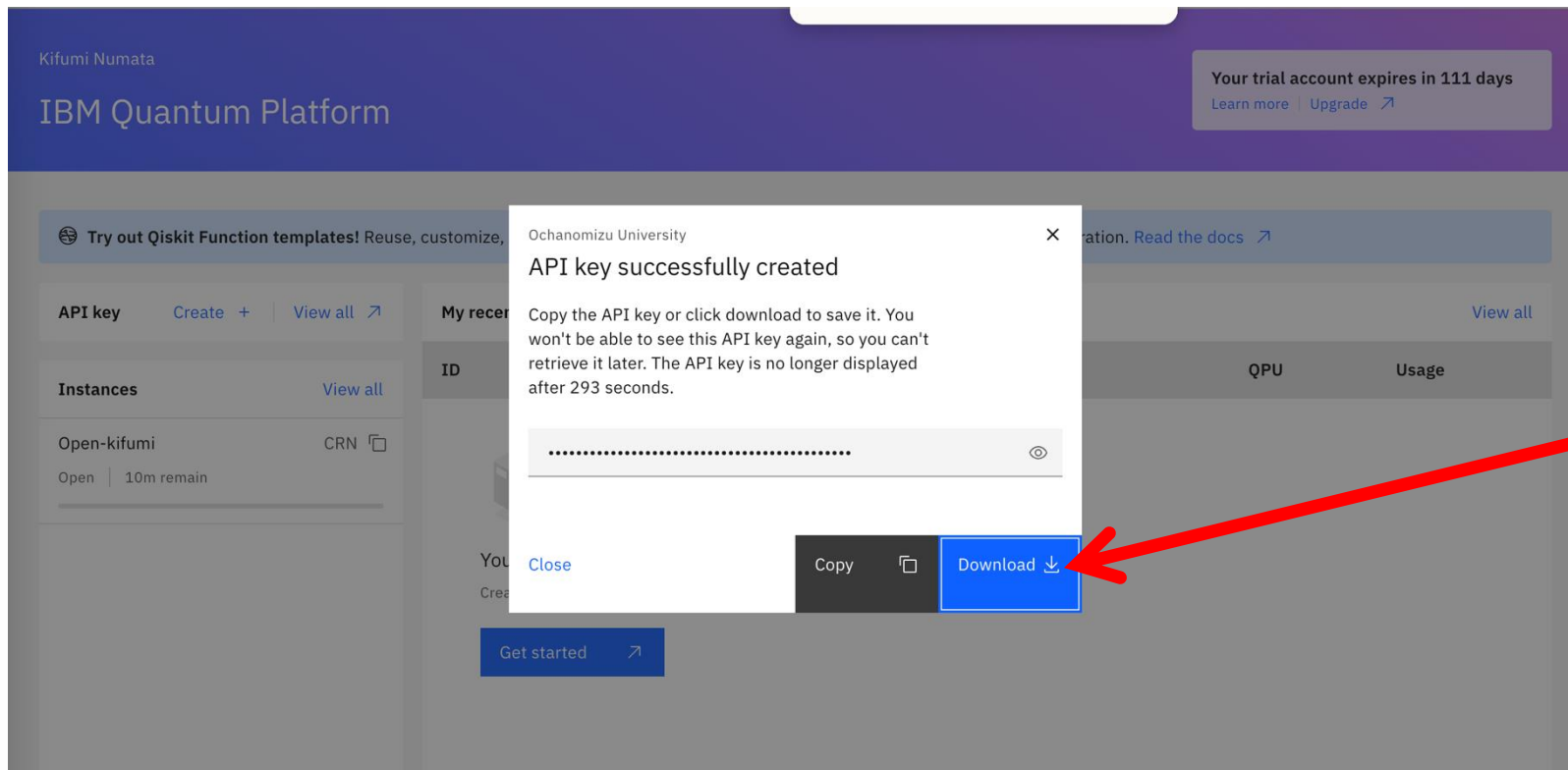
「Name」を入力します。（自分の名前など）



「Create」をクリックします。



API keyを「Copy」するか、または、「Download」をクリックし、保存します。



ダウンロードしたファイル「apikey.json」をダブルクリックして開くと、
API keyを確認できます。

JSON 生データ ヘッダー

保存 コピー すべて折りたたむ すべて展開 🔍 JSON を検索

```
name: "Open-kifumi"
description: "IBM Quantum API key"
createdAt: "2025-07-24T05:39+0000"
apikey: "[REDACTED]"
```

ダウンロードしたファイル「apikey.json」をダブルクリックして開くと、API keyを確認できます。



<https://quantum.cloud.ibm.com/> からCRNをコピーできます。

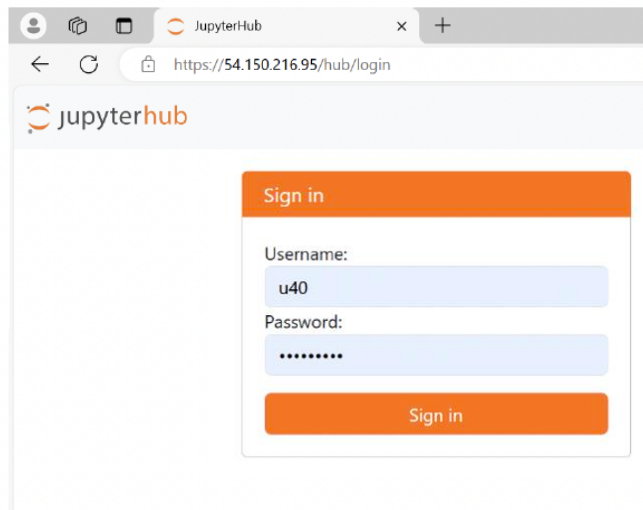
The screenshot shows the 'Qiskit Function' tab in the IBM Quantum console. It includes a 'Try out Qiskit Function' button, an 'API key' field with a 'Create +' link, and an 'Instances' section. A large red arrow points from the top left towards the 'Instances' section.

JupyterHubでの実行

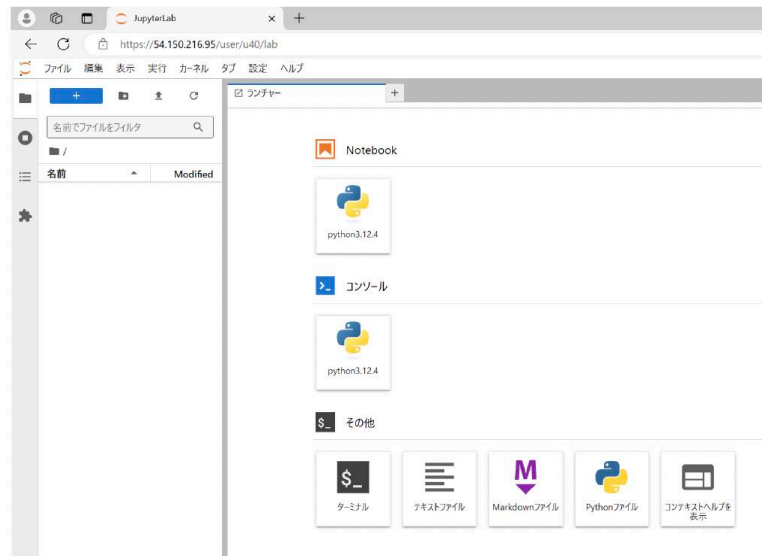
(1) Webブラウザ（ Edge、Safari、Chrome、Firefoxなど）で
<https://54.178.57.208/>にログイン。



(2) ユーザ名とパスワード（メールで配布）を
入力して、「Sign in」をクリック。



(3) この画面になったら成功です！



Kawasaki Campが終わった後、Qiskitを実行する場合

(1) Google Colabratory (<https://colab.research.google.com/>) を使う。
毎回、以下のコマンドを最初に実行する必要があります。

```
!pip install qiskit qiskit[visualization] qiskit-ibm-runtime qiskit-aer
```

```
!pip install qiskit-algorithms qiskit-nature scikit-learn
```

```
!pip install --prefer-binary pyscf
```

参照 : https://quantum-tokyo.github.io/introduction/get_started/colab.html

(2) qBraid (<https://www.qbraid.com>) を使う。

参照 : https://quantum-tokyo.github.io/introduction/get_started/qbraid.html

